

Lab05 - CI/CD Pipeline

1. Build ML Model

- Trained a Random Forest Model and saved it as model.pkl

2. Fast API Application

- UI on <http://127.0.0.1:8000/docs> and POST /predict

```
PS E:\Coursework\MLOps\iris-app> uvicorn app:app --reload
INFO: Will watch for changes in these directories: ['E:\\Coursework\\MLOps\\iris-app']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [150524] using StatReload
INFO: Started server process [178524]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:52538 - "GET / HTTP/1.1" 404 Not Found
INFO: 127.0.0.1:52538 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO: 127.0.0.1:56467 - "GET /predict HTTP/1.1" 405 Method Not Allowed
INFO: 127.0.0.1:52689 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:52689 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:59978 - "POST /predict HTTP/1.1" 200 OK
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [178524]
INFO: Stopping reloader process [150524]
PS E:\Coursework\MLOps\iris-app>
```

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "features": [5.1, 3.5, 1.4, 0.2]
  }'
```

Request URL

<http://127.0.0.1:8000/predict>

Server response

Code	Details
200	Response body <pre>{ "prediction": 0 }</pre> Response headers <pre>content-length: 16 content-type: application/json date: Mon, 09 Mar 2026 20:43:35 GMT server: uvicorn</pre>

3. Dockerize the Application

- Created the Dockerfile and .dockerignore
- `docker build -t iris-ml-api .`

```
PS E:\Coursework\MLOps\iris-app> docker build -t iris-ml-api .
[+] Building 47.2s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 386B
=> [internal] load metadata for docker.io/library/python:3.10-slim
=> [internal] load .dockerignore
=> => transferring context: 62B
=> [1/5] FROM docker.io/library/python:3.10-slim@sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aa
=> => resolve docker.io/library/python:3.10-slim@sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aa
=> [internal] load build context
=> => transferring context: 183.01kB
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:5c1fe7c14b73fbb5adb89a5f072ae9c2c10dfadc3ef7d4ac06d7d88f0bc701
=> => exporting config sha256:1ed81377ad619d67cc647c62b424b3374f6b7f881283ea95af86ea97d17ac393
=> => exporting attestation manifest sha256:33346908b7700df662eadb9002edfb13927d15c6dbf8e5f9c0a16e7ca7fb
=> => exporting manifest list sha256:3b14d6a2d883418e275efb4986c2c971db6e16fda46c9243b118c0dfdc619a71
=> => naming to docker.io/library/iris-ml-api:latest
=> => unpacking to docker.io/library/iris-ml-api:latest
PS E:\Coursework\MLOps\iris-app>
```

c. docker images

```
PS E:\Coursework\MLOps\iris-app> docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
iris-ml-api	latest	3b14d6a2d883	44 seconds ago	555MB
ml-api	latest	d6654ae136be	3 days ago	714MB
mongo	latest	d6b569590880	7 weeks ago	1.26GB
mongo-express	latest	1b23d7976f02	2 years ago	286MB

```
PS E:\Coursework\MLOps\iris-app>
```

d. `docker run -p 8000:8000 iris-ml-api`

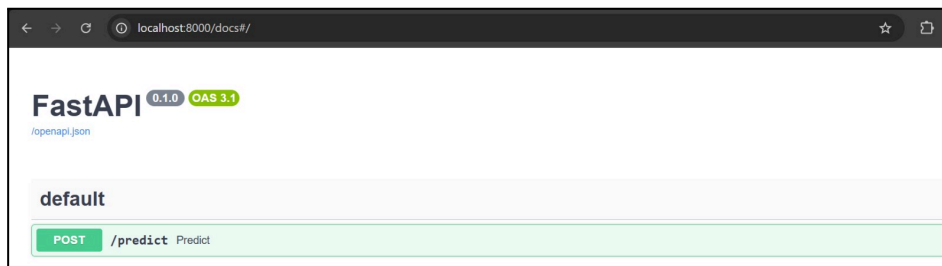
e. `docker ps`

```
PS E:\Coursework\MLOps\iris-app> docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b6d0ce13d0b0	iris-ml-api	"uvicorn app:app --h..."	15 seconds ago	Up 13 seconds	0.0.0.0:8000->8000/tcp	heuristic_chatelet

```
PS E:\Coursework\MLOps\iris-app>
```

f. UI on `http://localhost:8000/docs`



g. Testing the POST /predict



```
curl -X 'POST' \
  'http://localhost:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "features": [6, 2.7, 4.5, 1.5]
  }'
```

Request URL

http://localhost:8000/predict

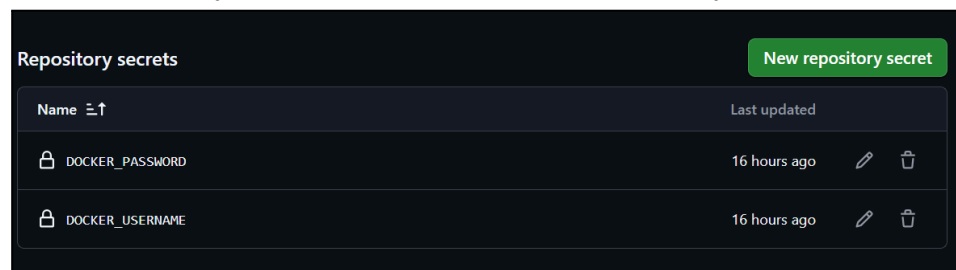
Server response

Code	Details
200	Response body <pre>{ "prediction": 1 }</pre> Response headers <pre>content-length: 16 content-type: application/json date: Mon, 09 Mar 2026 20:56:41 GMT server: uvicorn</pre>

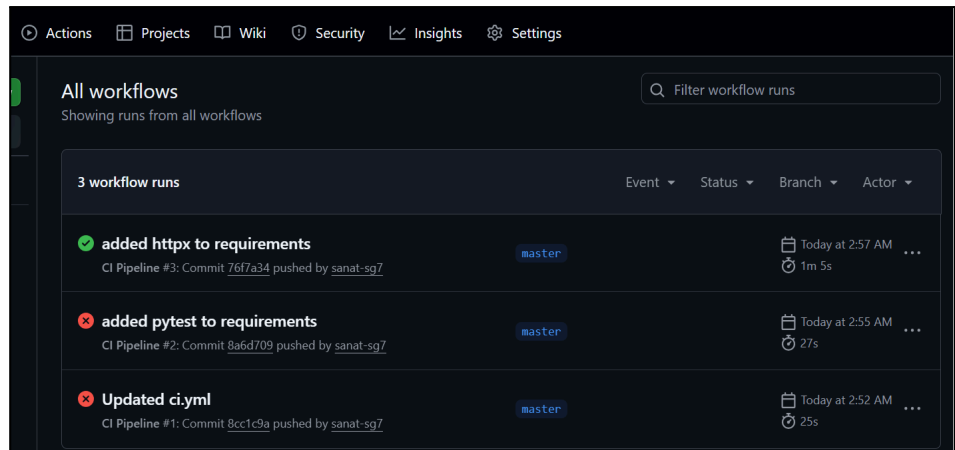
4. Converted the local project directory into a git repo
 - a. git init
 - b. git add .
 - c. git commit -m "Initial Commit"
 - d. git remote add origin https://github.com/<github username>/<repo name>.git
 - e. git push origin master

5. Implement GitHub Actions CI Pipeline

- a. Created .github/workflows/ci.yml file in the project directory
- b. On Github, Open the repo
 - i. Go to Settings > Secrets and Variables > Actions
 - ii. In the Repository Secrets section, add "New Repository Secret"



- c. Tested the CI Pipeline
 - i. git add .
 - ii. git commit -m "Testing the pipeline"
 - iii. git push origin master
- d. Any push, pull requests would trigger the CI Pipeline, which can be monitored by:
 - i. On Github, visit the Actions Tab of the repo and you can see the workflow running.
 - ii. If successful, a green tick would be seen in front of the workflow



- e. Any changes made into the repo, once pushed, would automatically ensure that a docker image is created of the latest version of the code and is pushed. Hence is called Continuous Integration and Continuous Development/Deployment.